

Venkata Obinni

Denton, Texas (Ready to Relocate) | Email: saiveeravenkataobinni@gmail.com | Phone: +1(940)-843-0820

LinkedIn: www.linkedin.com/in/obinni-data-engineer-analyst/

Data Engineer (Python, Spark, SQL, ETL, Snowflake, Kafka, Airflow, DBT) with 4 years of experience

Summary

- Data Engineer specializing in large-scale ETL pipelines, data platform architecture, and LLM-driven data extraction on cloud infrastructure.
- Built production data pipelines processing 500K+ records monthly using Python, PostgreSQL, Apache Airflow, and Spark, transforming unstructured data into structured analytics datasets.
- Engineered a multi-model LLM orchestration framework with 18 models and task-specific routing, achieving 90% reduction in manual data processing effort.
- Designed data canonicalization system using Vector DB, consolidating 289+ raw variants into 188 canonical groups for standardized taxonomy.
- Architected analytics engine with 8 modules and multi-dimensional cross-filtering using Spark, DBT, and PostgreSQL, delivering actionable career intelligence.
- Created parameter-driven, reusable pipeline architecture enabling zero-code expansion to new data domains and geographies.

Skills

Programming Languages: Python, Shell Scripting

Databases: SQL, PostgreSQL, Relational Databases

Big Data Processing Frameworks: Spark, PySpark

ETL and Data Integration Tools: ETL, Data Pipelines, AWS Glue

Cloud Data Platforms: AWS, S3, Google Cloud

Workflow Orchestration: Airflow, Apache Airflow

Data Warehousing: Dimensional Modelling, Star Schema, Snowflake, Databricks

Data Modelling: Schema Design, Data Pipeline Architecture

Streaming and Messaging Platforms: Kafka, Streaming

BI & Visualization: Power BI, Data Visualization

Professional Experience

Data Engineer

February 2022 – Present

Drizzly Technologies Inc. (SkillZelo.com)

Vancouver, Canada (Remote)

SkillZelo provides job-market analysis using massive datasets. SkillZelo helps Universities, students, employees, and employers with job market analysis and job search support tools.

Project/Product: SkillZelo

- Working as Data Engineer on a SaaS-based job market analysis and career intelligence platform within a 4-member engineering team at a startup, processing 500K+ job postings monthly across 400+ job titles.
- Built end-to-end ETL pipelines using Python and Apache Airflow to ingest, parse, and normalize thousands of job postings into structured datasets stored in PostgreSQL, forming the platform's core data infrastructure.

- Developed LLM-based extraction pipelines leveraging a multi-model architecture with 18 models to parse unstructured job descriptions into structured fields covering skills, duties, qualifications, salary, and experience data.
- Engineered a two-layer canonicalization system using Vector DB similarity matching to consolidate 289+ raw skill variants into 188 canonical groups across technical, functional, and soft skill categories.
- Designed the aggregation engine powering 8 analytics modules with multi-dimensional cross-filtering across salary, experience, geography, and work types using DBT-managed SQL transformations.
- Created a parameter-driven, reusable pipeline architecture enabling new market analyses for any role/country combination without writing new code, reducing launch time from weeks to a single data load.
- Architected the LLM execution orchestration framework supporting 7+ configurable execution types with parallel batch processing, per-model cost tracking, and automatic/manual trigger modes.
- Built the profile-to-market matching system and 2N+2 package generation pipeline to produce personalized career deliverables by mapping job seeker profiles against live market data.
- Implemented data quality engineering practices including validation, deduplication, and completeness tracking, transparently reporting coverage gaps across all analytics modules.
- Optimized LLM workload costs through task-specific model routing, batch processing, and parallel execution management, achieving 90% reduction in manual data processing effort.
- Technical Environment: Python, PostgreSQL, Apache Airflow, DBT, Spark, PySpark, Pandas, AWS (S3, Lambda, Glue), Docker, Jenkins, Git, GitHub, Elasticsearch, Vector DB, LLM APIs (18 models), Parquet, Power BI, Linux, Shell Scripting, Jira, Confluence

Education

Master's in data science
University of North Texas

January 2024 – December 2025
Deton, TX, USA

Bachelor of Technology in Mechanical Engineering
Sastra Deemed University

June 2019 – May 2023
Tamil Nadu, India

Skills Continuation....

Big Data & Processing: Spark, Databricks, Delta Live Tables, Parquet, Data Lake

Cloud & Serverless: AWS Lambda, Step Functions

More Databases: Elasticsearch, NoSQL

Data Engineering: ETL, Data Ingestion, Data Transformation, Data Processing, Data Extraction

Data Quality & Governance: Great Expectations, Data Validation, Data Quality

ML & AI: Machine Learning, LLMs, Pandas

APIs: REST API, API Development

DevOps & CI/CD: Docker, Jenkins, CI/CD, Git, GitHub

Enterprise Platforms: Salesforce, CRM, ERP, SAS

Query & SQL: Databricks SQL

OS & Scripting: Linux, Shell Scripting

Tools: Jira, Confluence